

Behavioral Economics: Principles, Procedures, and Utility for Applied Behavior Analysis

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Abstract

Behavioral economics is the merger of microeconomic principles with the procedures and principles of the experimental analysis of behavior. This review describes two branches of behavioral economic research: variables affecting decision making and consumer demand. Each branch is discussed with an eye toward identifying behavioral economic principles that will be of utility to those tasked with designing effective behavioral interventions.

Keywords: behavioral economics, delay discounting, demand, price, applied behavior analysis

Behavioral economics, as the name implies, is a hybrid area of research. One form of this hybridization is the use of principles, procedures, and measures associated with behavior-analytic research (e.g., direct observations of steady-state operant behavior). However, Googling “behavioral economics” reveals a second hybrid that will be less familiar to readers of this journal, but more familiar to economists. This second form of behavioral economic research represents an infusion of decision-making research findings into mainstream economics. Most of these findings have been contributed by cognitive psychologists such as Kahneman and Tversky (1979), working in the area of prospect theory. Discoveries made by these researchers have had a profound influence on economists because they demonstrate reliable deviations from expected utility (maximization) theory – an assumption upon which much economic theorizing was based. This influence was recognized by the awarding of the 2002 Nobel Prize in economics to Daniel Kahneman, the father of prospect theory.

The methods of research used by prospect theorists rely primarily on asking participants to make a single choice between hypothetical prospective outcomes (e.g., to choose between a sure thing and a risky outcome). As in most psychological research, choices are averaged across individuals and subjected to statistical analyses. No doubt, some readers of this journal will be critical of some components of these research methods (e.g., the practice of measuring self-reports of what one might do instead of measuring choices between real outcomes). Such critiques are worth empirical investigation (e.g., Johnson & Bickel, 2002), but it should be recognized that some very important choices made by humans are prospect choices; that is, one-time choices where the individual weighs prospective outcomes before they are experienced. For example, when deciding to accept or reject the offer of a promotion at work, the individual will make a single choice (rather than a sequence of choices continuing until choice stabilizes). Although it would be interesting to determine if prospect choices represent steady-state choices made after behavior has reached an equilibrium with the prevailing contingencies, studying choice in this way will take it out of the context of choosing between prospective outcomes, and may tell us little about some very important decisions that humans are called upon to make. Consider, for example, the prospect choice that members of the U.S. congress were being asked to make at the time this paper was written – whether or not to accept the Treasury Secretary’s advice to back failing Wall Street financial institutions with \$700 billion of taxpayer money. Unprecedented choices are choices between prospective outcomes, and these choices are worthy of a thorough behavioral analysis.

An obvious place to begin such an analysis would be to study how prior experiences (i.e., history) affect prospect choices (e.g., experiences with outcomes similar to those that will be chosen

between on a prospect basis). It goes without saying that the decisions made by members of the U.S. Congress will be affected by their prior experience with similar economic crises (e.g., the financial bailout and return to solvency of the Chrysler Corporation in the 1980s). Also obvious is that a rat's steady-state pattern of choice is a product of its history of experiencing the consequences of the choices it makes (pardon us for comparing rats to members of Congress). No one will be surprised to see the rat's first choices in a new condition of the experiment, analogous to the prospect choices made by humans, affected by prior reinforcement history (see Lattal & Neef, 1996, for a review). To the extent that the new choice context (e.g., an economic bailout of Wall Street) resembles the old one (e.g., the bailout of Chrysler), we would expect the rat (or the member of Congress) to behave as he or she did in the past if that pattern of choice had been selected by consequences.

This is all seemingly obvious and, perhaps for that reason, has not inspired much research into how history of reinforcement affects prospect choice. However, as noted by Lattal and Neef (1996), therapeutic interventions may be thought of as artificially arranged conditioning histories and one measure of the utility of such interventions is the extent to which behavior is effected in new settings (e.g., prospect choices). Given the importance of prospect choices and the impact of prospect theory on psychological and economic research, there may be something important to be learned about how best to arrange a history of operant contingencies to positively impact prospect choices. Likewise, knowledge of the regularities of human choice identified by prospect theorists may be of some interest to those arranging contingencies of reinforcement. Consider a central finding of prospect theory, that aversive events have a greater impact on choice than benefits. There is little doubt that increased sensitivity to prospective losses describes group-averaged prospect choices made by humans (e.g., Kahneman & Tversky, 1984), but it is unknown if this is a phylogenetic tendency which may be affected by an extended history of ontogenetic experiences (see Magoon & Critchfield, 2008 for a review of the literature on this topic). A more thorough understanding of this tendency, and historical experiences that may affect it, may be useful in understanding a variety of behavioral events of importance in applied settings.

While one-time prospect choices are often important, a good number of choices are clinically significant because they are repeated to the point of "self-destruction" (e.g., substance abuse and pathological gambling). Such repeated choices represent steady-state patterns which impact long-term health and wealth. We now turn to advances in behavioral economics that have provided insights into such self-destructive choices, and which provides a bridge between prospect theory and the study of steady-state operant behavior.

Delay Discounting

Delay discounting is often referred to as a behavioral process by which an organism discounts the value of a reinforcer because its delivery is delayed. Such language is a shorthand - delay discounting merely describes a temporally extended pattern of choices from which the researcher may deduce how the efficacy of a reinforcer declines as the delay to its delivery increases. Considerable resources have been allocated to studying delay discounting with nonhuman subjects and the results are impressive. Using steady-state procedures in which animal subjects choose between small-immediate and large-delayed reinforcers, researchers have quantified how the value of a reinforcer declines as it becomes increasingly delayed. In what must be regarded as an amazing example of cross-species regularity, for rats, pigeons, and monkeys, reinforcer value declines according to a hyperbolic decay function as it is delivered following longer delays (e.g., Mazur, 1987; Richards, Mitchell, de Wit, & Seiden, 1997; Woolverton, Myerson, & Green, 2007). This is illustrated in Figure 1. Across species, the steepness of the curve varies, but the same hyperbolic equation describes the animals' steady-state choices quite well.

There are several interesting things to point out about the hyperbolic delay-discounting curve shown in Figure 1, but before we do, we want to make two final points about prospect theory and how it has affected behavioral economics. First, when humans make prospect choices between hypothetical immediate and delayed rewards, their pattern of choices conforms to the same hyperbola describing animal choice (e.g., Green, Fry, & Myerson, 1994; Grossbard & Mazur, 1986; Kirby, 1997; Madden, Bickel, & Jacobs, 1999; Rachlin, Raineri, & Cross, 1991; Simpson & Vuchinich, 2000; although see

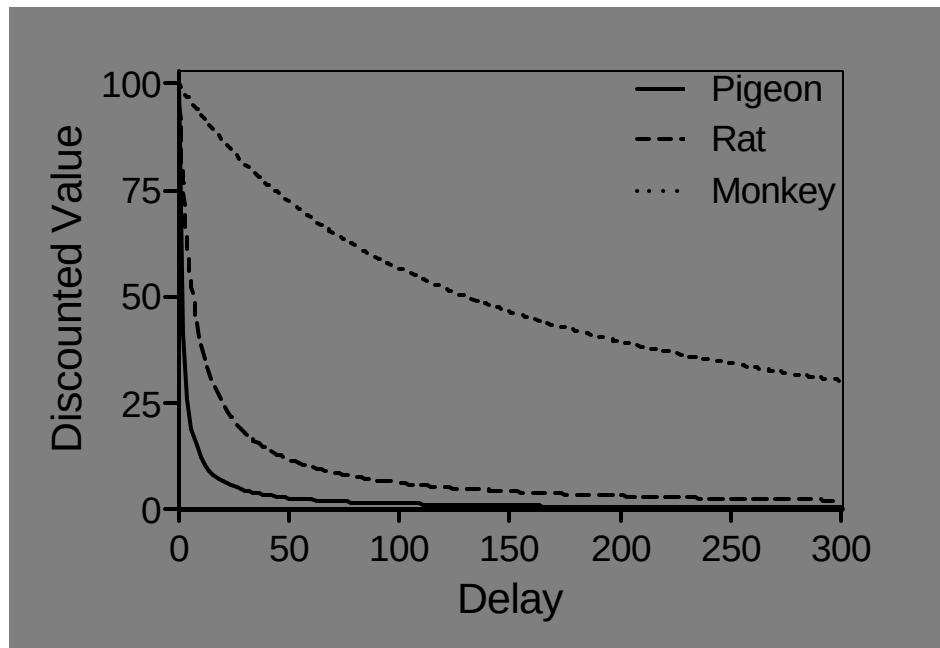


Figure 1. Cross-species delay discounting curves as described by the hyperbolic delay function. Discounted value of the larger-later reinforcer is plotted on the y-axis, as a function of increasing delays (x-axis). Hyperbolic discounting is represented by the solid curve for pigeons, by the dashed curve for rats, and the dotted curve for monkeys. Discounting curves were drawn by setting the discounting rate at those obtained empirically with pigeons, rats, and monkeys by Mazur (1987), Richards, Mitchell, de Wit, and Seiden (1997), and Woolverton, Myerson, and Green (2007), respectively.

Green & Myerson, 2004, for evidence that for humans the curve may be more hyperbola-like than strictly hyperbolic). This finding has been obtained regardless of whether the outcomes are real or hypothetical (e.g., Johnson & Bickel, 2002; Madden, Begotka, Raiff, & Kastern, 2003) or whether delayed gains or losses are considered (e.g., Murphy, Vuchinich, & Simpson, 2001; Odum, Madden, & Bickel, 2002). These findings represent an important use of prospect-choice methods to determine the generality to humans of hyperbolic delay discounting observed in the patterns of steady-state behavior of nonhuman animals.

Second, economists have taken note of delay discounting research because the hyperbolic shape of the discounting function is not predicted by normative economic theory. The latter holds that the value of a delayed outcome should be discounted in an exponentially compounding fashion over time (e.g., Samuelson, 1937); one might refer to this as “compounding disinterest.” This assumption of economic theory underlies the compound interest rates used by banks; if consumers discount delayed money at a compounding rate, then to secure their deposits the bank must offer to grow their money in a compounding fashion that offsets exponential delay discounting. That humans and three other animal species all violate normative economic theory is difficult to ignore. But of equal interest to economists is

that hyperbolic delay discounting appears to help explain some forms of irrational behavior that make little sense from a normative economic perspective.

Irrational Choice. Exponential delay discounting predicts that, all else being equal, preference should be rational in the sense that it remains constant over time. According to this definition, a rational cigarette smoker who decides to quit smoking (and spends considerable money on nicotine replacement products, behavior therapy, etc.) would remain committed to this decision over time. Likewise, the rational consumer who decides to pay off his or her credit card debt would resist daily temptations to impulse buy so that substantial payments may be made at the end of each month. However, anyone who has attempted to quit smoking, pay down a debt, eat healthy, or exercise knows that the best of intentions are soon lost.

Perhaps these real-world examples have ignored the “all else being equal” clause – perhaps something changed over time, and that something caused the individual to “irrationally” start smoking or spending again. Unfortunately, from the perspective of normative economic theory, laboratory studies with both humans (Green & Myerson, 2004) and nonhumans (Ainslie & Herrnstein, 1981; Green & Estle, 2003; Green, Fisher, Perlow, & Sherman, 1981) that control for such “all else not being equal” accounts, have confirmed our suspicion that choice is not consistent over time.

Interestingly, these choice inconsistencies are predicted by the deeply bowed shape of the hyperbolic discounting function shown in Figure 1. To illustrate this, Figure 2 shows two vertical bars corresponding to two reinforcers; one twice the size of the other. Along the x-axis we have plotted the time separating a choice and the delivery of the reinforcer. At time T1, the smaller reinforcer is immediately available (no temporal distance separates T1 from the smaller bar), while the larger reinforcer is delayed. At time T2, both reinforcers are delayed, with the difference in delivery time at T2 being the same as at T1. Also shown in Figure 2 are two hyperbolic delay discounting curves, both using the same rate of discounting. Each discounting curve maps the value of the reinforcer as it is delivered following a range of delays. If we assume that an individual will choose the reinforcer with the greater

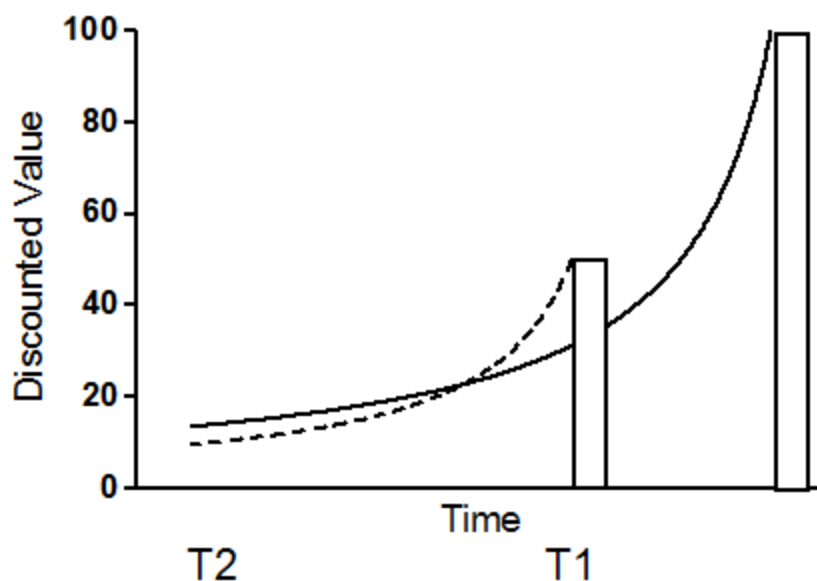


Figure 2. Intertemporal choice as predicted by the hyperbolic delay discounting equation, given identical rates of discounting for two reinforcers. Reinforcers are represented by vertical bars, with height indicating magnitude. Time

separating choice and delivery of a reinforcer is plotted on the x-axis, with $T1$ and $T2$ corresponding to times at which an organism might choose between a smaller-sooner and a larger-later reinforcer. The discounted value of a reinforcer at a given point in time is plotted on the y-axis. The curves to the left of each reinforcer (solid curve associated with the larger reinforcer, dotted curve associated with the smaller reinforcer) are hyperbolic discounting functions that map discounted reinforcer value over a range of delays. At any given point in time, the reinforcer with the higher value curve is preferred. The point at which the curves intersect indicates a reversal of reinforcer preference.

discounted value, then at time $T2$ the individual should choose the large-delayed reward (the discounting curve for the large-delayed reward is higher than that for the small-immediate reward). Thus, our consumer will judge the delayed benefits of paying down his debt as exceeding the less-delayed benefits of another purchase. Under these conditions, the consumer cuts up his credit cards and takes out a debt-consolidation loan. Unfortunately, something dreadful happens as we proceed from $T2$ to $T1$: the relative discounted values of the reinforcers are switched as an immediate temptation (e.g., an expensive meal, a plasma-screen TV) is encountered. Under these conditions, the consumer may take advantage of the many instant credit offers that retailers use to entice us to make an impulsive choice. While this irrational choice is not predicted by the economist's exponential discounting process (exponential discounting curves do not cross), it is by hyperbolic delay discounting.

The inter-species generality of hyperbolic delay discounting seems to suggest that we are doomed by our phylogenetic heritage to make irrational inter-temporal choices. How could natural selection have prepared us so poorly? Why aren't we rational maximizers who, once we commit to a large-delayed outcome at $T2$, stick with that choice as we approach $T1$? According to the ecological rationality hypothesis, choosing immediate over delayed reinforcers may have offered organisms a selective advantage in a natural environment filled with predators, competition, unpredictable food supplies, and scarce mating opportunities (e.g., Stevens & Stephens, in press). These tendencies, well honed by natural selection, are not well suited to the unnatural context in which consumer choices are frequently made: a context in which consumables never imagined by our foraging ancestors are now readily available if we are just willing to sacrifice our long-term health or wealth in order to consume them. Does our phylogenetic history (dealing us a hand of hyperbolic delay discounting) doom us to make irrational choices in much the same way that a species well suited to its home niche undergoes extinction when radical changes to the niche occur? Perhaps not.

The last decade has revealed some interesting findings that speak to this question. First, although hyperbolic delay discounting is common to all of the species tested thus far, there are considerable inter- and intra-species differences in the rate at which delayed reinforcers are discounted (e.g., Anderson & Woolverton, 2005; Green, Myerson, Holt, Slevin, & Estle, 2004; Madden, Smith, Brewer, Pinkston, & Johnson, 2008; Mazur & Biondi, 2009). These differences matter because lower rates of delay discounting predict reduced irrational inter-temporal choices. This is illustrated in Figure 3, which shows the same choice alternatives as in Figure 2, but with two different rates of delay discounting. The top curve illustrates a lower rate of hyperbolic delay discounting than the bottom curve. At this low rate of discounting, the discounted value of the large-delayed reinforcer always exceeds that of the small-immediate reinforcer. Thus, if our indebted consumer discounts delayed rewards at this low rate, he will cut up his credit cards at $T2$ and resist the temptation of the plasma-screen TV at $T1$ because the discounted value of a debt-free future always exceeds the undiscounted value of the TV. However, at a higher discounting rate (lower curve) the consumer will behave irrationally, seeing clearly the value of a debt-free future at $T2$ and reversing his commitment at $T1$.

Consistent with the hypothesis that the tendency to heavily discount the value of delayed reinforcers may underlie irrational choices, a large body of evidence suggests that individuals diagnosed

with drug and alcohol addictions and pathological gambling discount delayed prospect rewards at a higher rate (lower discounting curve in Figure 3) than matched control participants with no history of addictions (see reviews by Petry & Madden, in press, and Yi, Mitchell, & Bickel, in press;). Some evidence suggests that rats that demonstrate high rates of delay discounting are more likely to self-administer drugs of abuse (see review by Carroll, Anker, Mach, Newman, & Perry, in press), and Madden, Ewan and Lagorio (2007) have argued that preference for gambling is a mathematical necessity of high rates of hyperbolic delay discounting.

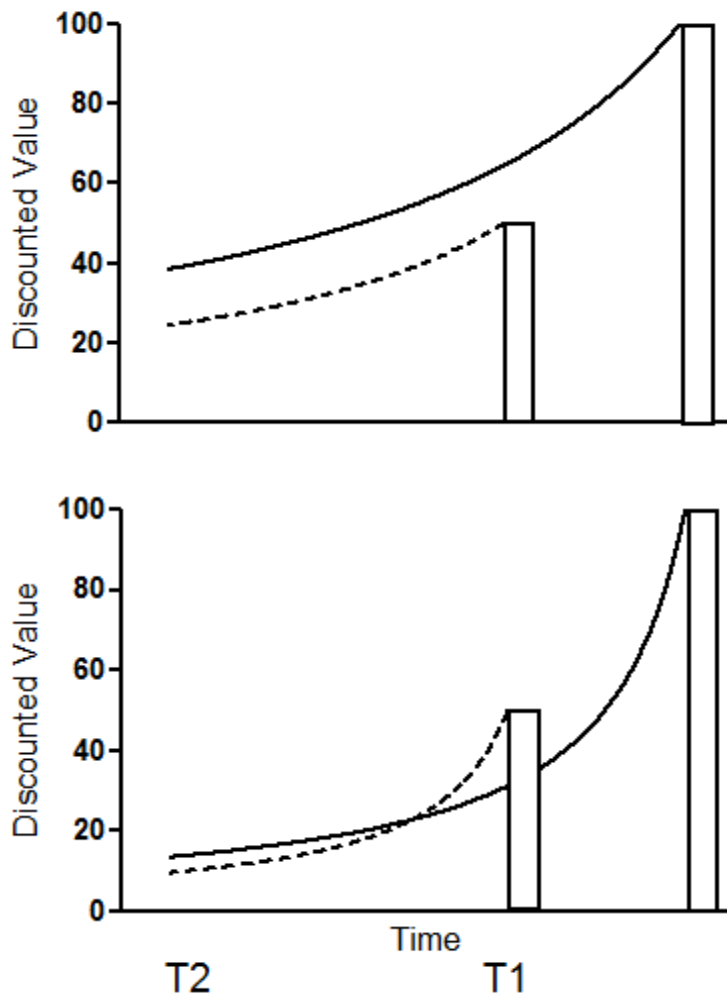


Figure 3. Comparison of intertemporal choice as predicted by the hyperbolic delay discounting equation, given different rates of discounting. The top panel illustrates the same reinforcers as in Figure 2, except that they are discounted at relatively lower rates. The bottom panel is identical to Figure 2, displayed as a means of comparison. All features of Figure 3 are identical to those in Figure 2.

A question of obvious applied utility is how to move an individual from the lower to the upper discounting curve in Figure 3. Said another way, how does one teach an individual to better tolerate delays to reinforcement? In studies, Mazur and Logue (1978) and Schweitzer and Sulzer-Azaroff (1988) demonstrated techniques for shaping delay tolerance in pigeons and children, respectively (see related studies by Dixon, Rehfeldt, & Randich, 2003; Logue, Rodriguez, Peña-Correal, & Mauro, 1984; Vollmer, Borrero, Lalli, & Daniel, 1999). Pigeons in the Mazur and Logue study first learned to select a large over

a small food reinforcer when both were delivered following a 6-s delay. When the pigeons reliably selected the larger reinforcer, the delay to the smaller one was very gradually decreased over the course of 1 year until the pigeons were choosing between a small-immediate and a large-delayed (6-s) reinforcer. In this final condition, these pigeons, and children in the study by Schweitzer and Sulzer-Azaroff, who were given a similar (though briefer) conditioning history, more frequently chose the large-delayed reinforcer when compared with a control group given no such training history. When Mazur and Logue's pigeons were re-assessed approximately 1 year later, they continued to demonstrate good tolerance for delay (Logue & Mazur, 1981). These important findings suggest that individuals can learn to reliably select a larger-later over a smaller-sooner outcome. However, unanswered by these findings are questions concerning generalization to novel settings, longer delays, and different reinforcers. Successful generalization of this sort might be characterized as the acquisition of a generalized delay tolerance. Given the evidence suggesting that delay intolerance (i.e., high-rate delay discounting) is predictive of drug taking (e.g., Perry, Larson, German, Madden, & Carroll, 2005), relapse to drug taking (e.g., Perry, Nelson, Carroll, 2008), and poor outcomes in substance-abuse treatment (e.g., Yoon, Higgins, Heil, Sugarbaker, Thomas, & Badger, 2007), identifying effective techniques for training delay tolerance is an important goal of basic and applied research in behavioral economics.

As a first approximation of this research line, Vollmer et al. (1999) taught 2 children with disabilities who engaged in aggression to use a functional communication card (a mand) to request a tangible item. In an impulsivity test phase, children could obtain a small-immediate consumable item by behaving aggressively, or they could mand for a large-delayed consumable. When mands were acknowledged, either by the experimenter placing a hand in a bag of chips throughout the delay or by activating a timer observable by the child, what might be termed "impulsive aggression" was decreased. For 1 child, the experimenters gradually increased the delay to the large-delay reinforcer and high percentage "self-controlled" manding was maintained. For this child, the timer was eventually incorporated into the daily routine to promote self-control and decrease impulsive aggression.

We now turn our attention to a second area of behavioral economic research: the study of consumer demand in individuals.

Consumer Demand and Price

The branch of economics known as microeconomics is concerned with the relation between consumers (i.e., those who purchase goods) and suppliers (those who sell the goods). Consumer demand refers to the purchasing activities of the consumer. How much of the good will the individual purchase when it is available at a very low price? How much will continue to be purchased as the price of the good increases?

A number of basic laboratory researchers have conceptualized the lever-pressing or key-pecking behavior of their animal subjects as the behavior of a consumer (e.g., Hursh, Raslear, Shurtleff, Bauman, & Simmons, 1998; Madden, Dake, Mauel, & Rowe, 2005). When the experimenter manipulates the schedule of reinforcement, he/she is acting as a supplier who sets the price of the commodity offered in the marketplace. In these experiments, price is defined as it is in economics – as a cost-benefit ratio termed unit price. The unit price of a good specifies the price paid for a standard unit of the reinforcer and can be increased by increasing costs (e.g., lever presses per reinforcer) or by decreasing the benefit (e.g., reinforcer magnitude).

If one conceptualizes the animal's operant behavior as spending (i.e., the allocation of limited resources to the procurement of goods), then the experimenter will diminish his/her profits (e.g., tick marks on a cumulative recorder) by making the reinforcer available at a low price (e.g., reinforcing every

response). This is illustrated in the upper panel of Figure 4. When the experimenter sets the price at one lever press per food pellet (unit price = 1.0), consumer spending (i.e., lever presses per session) is low. When the unit price is increased (e.g., by requiring more responses per reinforcer), the subject increases responding until a price is reached at which responding declines. This general effect is frequently referred to as “ratio strain.” Behavioral economists refer to the specific price at which peak responding is reached as P_{max} (shown as the vertical line in both panels of Figure 4). At price increases exceeding P_{max} , consumer spending declines.

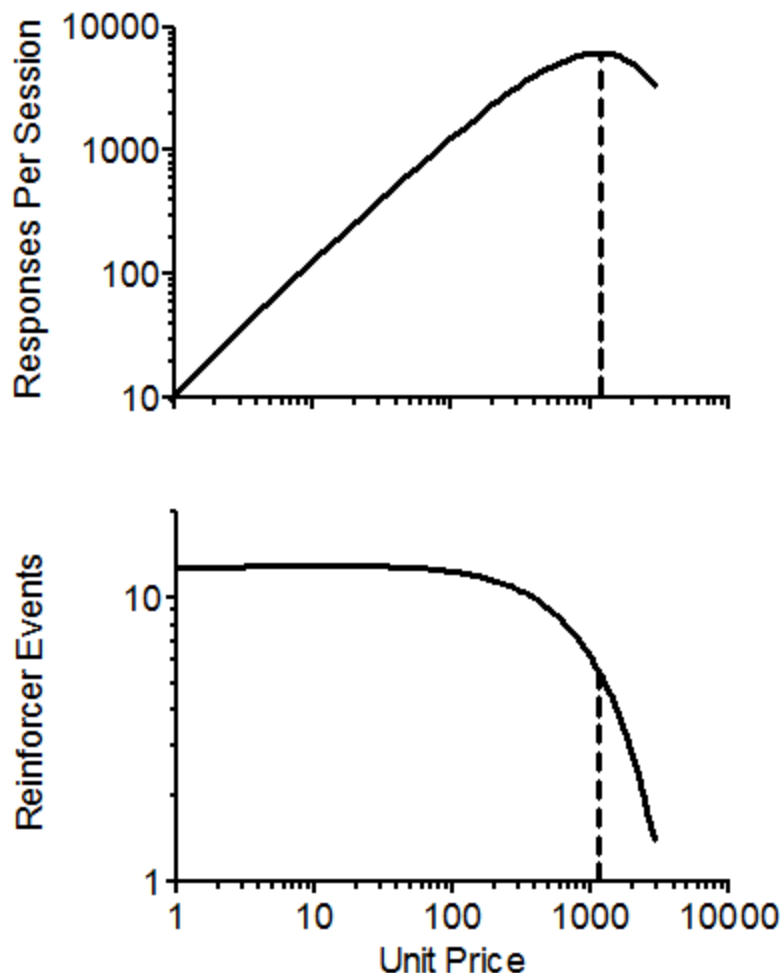


Figure 4. Hypothetical consumer spending and consumption demand curves (top and bottom panels, respectively). The top panel shows the relation between the number of responses emitted per session for a commodity as a function of increasing price. The bottom panel illustrates the effects of increasing unit price on consumption of reinforcer events. In both panels, the vertical dotted line represents P_{max} , the price at which peak responding is reached.

The shape of the consumer spending curve shown in the upper panel of Figure 4 characterizes rats' lever pressing for food, water, fat, electronic brain stimulation, and a variety of drugs (e.g., Bickel, DeGrandpre, Higgins, & Hughes, 1990; Collier, Hirsch, & Hamlin, 1972). It also characterizes pigeons' key pecking for food (e.g., Madden et al., 2005), humans' plunger pulling maintained by cigarette puffs (e.g., Bickel & Madden, 1999), and spending on grocery items in the natural economy (e.g., Oliveira-Castro, Foxall, & Schrezenmaier, 2006). Consider your own spending on gasoline in 2008 when gas prices fluctuated from a low of less than \$1.50 to a high of over \$4.00 per gallon in the U.S. As prices

increased, so did the amount of money spent on gas per month, moving from left to right up the spending curve shown in Figure 4. When prices later fell, spending slid back down the left portion of the curve. If this description strikes the reader as odd, this is probably because we typically talk about gasoline consumption rather than money spent on gasoline.

As shown in the lower panel of Figure 4, as the price of a commodity like gasoline increases, consumption of that commodity declines (e.g., Congressional Budget Office, 2008). This straightforward inverse relation between price and consumption is called the demand law and has been confirmed in countless laboratory experiments (e.g., Foltin, 1991; Hursh et al., 1988; Johnson & Bickel, 2006). For a demand curve to provide a complete picture of consumption of a reinforcer across a wide range of prices, it is important that there be no arbitrary limits placed on daily consumption (this is a point often missed by those attempting to use behavioral-economic methods and measures). In typical behavioral-economic experiments, the reinforcer is available throughout long-duration sessions (e.g., 12 hr). In a long-duration session, the subject has ample opportunity to consume large quantities of the reinforcer such that peak consumption (at low prices) is determined by satiety, rather than by an experimenter-imposed cap. In Figure 5, that peak is 80 reinforcers per session. Measuring peak consumption is critically important when the experimenter is interested in measuring sensitivity to price increases. For example, if peak consumption is artificially capped at 20 reinforcers (open symbols in Figure 5), then the effects of a price increase will not be evident until a price is reached at which consumption decreases to below 20 reinforcers. Had the experimenter conducted long-duration sessions in which consumption was not capped, then he/she would be in a position to measure consumption decreases between 80 and 20 reinforcers. With a cap in place, the experimenter erroneously concludes that a wide range of price increases have no effect on consumption. Because consumption of naturally occurring reinforcers is often not artificially capped, conducting long-duration sessions in behavioral-economic experiments provides a more appropriate model of behavior occurring in applied settings. As it turns out, exigencies of clinical contexts often do not permit evaluations of unrestricted spending and consumption. That is, it would be impractical to determine the value of a highly preferred tangible item by permitting (requiring) the consumer to “work all day.” The notion that restricting maximal consumption can impact determinations of price changes should, however, be considered by applied researchers attempting to bridge the basic nonhuman and human laboratory research on behavioral economics.

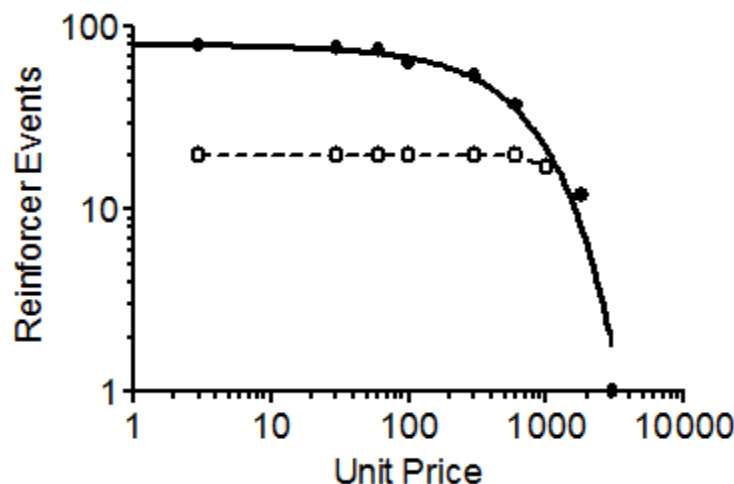


Figure 5. Hypothetical consumption demand curves generated when number of reinforcers that can be earned during a session are capped (dotted curve) and not capped (solid curve). Filled symbols on the top curve and unfilled symbols on the bottom curve represent the number of reinforcers earned per session.

Recently, applied behavior analysts have begun to use the demand law to conceptualize ways of reducing problem behavior. Focusing on consumption, Piazza, Roane, Keeney, Boney, and Abt (2002) reduced 3 children's pica (the dangerous ingestion of inedible objects) by increasing the effort required to obtain pica items. The price of pica items was increased by moving the items below the waist for 1 child (she was observed to engage in pica only when items were above her waist) or enclosing pica items in lidded containers for 2 children. Thus, the cost component (numerator) of the cost-benefit ratio was increased while the benefit component (i.e., the pica items themselves) was unchanged; the end result was a price increase and a decrease in consumption of pica items. Pica was further reduced when access to alternative leisure items was available at a low price. Other price manipulations could have been implemented by increasing search time (e.g., hiding the pica items) or decreasing the benefit component (e.g., reducing the size of the pica items). Similar manipulations of effort have been applied to self-injurious behavior (e.g., Hanley, Piazza, Keeney, Blakeley-Smith, & Worsdell, 1998; Van Houten, 1993) and more recently to response class hierarchies (Shabani, Carr, & Petursdottir, 2009). Shabani et al. manipulated the effort (force required) to complete one of three button presses. When all responses were available for reinforcement, participants consistently selected the alternative that required the least effort. When only two of three responses were available (e.g., the one requiring the most force and the one requiring the least force), participants consistently responded on the alternative that required comparatively less effort. In short, the participants purchased the cheapest commodity available, when cost was manipulated by requiring differing levels of effort.

Similar price-based principles have been shown to affect cigarette smoking. Madden and Bickel (1999) demonstrated that increasing the dollar price of cigarette puffs led to a decrease in cigarette consumption in a laboratory setting, producing demand curves similar to those observed with animal subjects for food. In a behavioral economic reanalysis of several studies, DeGrandpre, Bickel, Hughes, & Higgins (1992) found that in some cases nicotine intake (i.e., consumption) decreased when smokers were switched from their usual high nicotine yield cigarettes to low-nicotine cigarettes. This is predicted from the demand law because decreasing nicotine yield increases the unit price of nicotine (i.e., nicotine obtained per puff). Where nicotine intake decreased, it tended to do so at high unit prices. In response to this price increase, cigarette consumption increased, as in the consumer spending curve shown in Figure 4. This, of course, represents an increased health risk to the smoker.

On a larger scale, some correlational studies have suggested that a high density of tobacco outlets may be associated with higher per capita rates of cigarette smoking (e.g., Chuang, Cubbin, Ahn, & Winkleby, 2005). By minimizing the cost of traveling to a store at which cigarettes may be purchased, the unit price of cigarettes is kept low. From the demand law, one would predict that increasing these travel costs would, to some extent, decrease smoking. The extent to which demand for a commodity is affected by price is the next topic for discussion.

Sensitivity to Price Increases. Economists are interested in measuring sensitivity to price changes. For example, if we could quantify sensitivity of demand for cigarettes to price fluctuations, then we would be in a position to predict how much a sin tax on cigarette sales might decrease smoking. Likewise, a company owner may wish to know how sensitive is demand for his/her product to price changes so that the suggested retail price of the product could be set so as to maximize profits (i.e., approach P_{max} in Figure 4). Setting the price too high (going beyond P_{max}) would be a costly mistake.

This sensitivity to changing price is referred to as elasticity of demand. If a 1% price change results in less than a 1% change in consumption of a commodity, then demand for that commodity is said to be inelastic. Importantly, consumer spending increases with price increases for as long as demand remains inelastic. Inelastic demand occurs on the portion of the demand and consumer spending curves to

the left of P_{max} (see Figure 4). At some price increase, consumer's demand becomes elastic. When demand is elastic, a 1% price increase produces a greater than 1% decrease in consumption and consumer spending decreases with the price increase.

If the applied behavior analyst's goal is to maximize appropriate behavior (analogous to consumer spending), then it would be important to know how elastic the client's demand is for the reinforcer being offered. Although it will strike some as crass to suggest it, we believe that some of the work of applied behavior analysts can be conceptualized as similar to that of a company owner seeking to maximize income. From this perspective, the job of the applied behavior analyst is similar to that of the supplier of consumer goods: bring a product to market that clients will purchase even when the price of that commodity increases and other alternatives are available at a cheaper price (e.g., reinforcers obtained for engaging in problem behavior). Applied behavior analysts might begin by offering a reinforcer at an initially low price, then gradually increasing the price when the client is reliably responding and consuming the reinforcer. The applied behavior analyst or practitioner should be very comfortable with this type of arrangement as it characterizes the "schedule thinning" that occurs so frequently. By increasing, for the example, the delay to a break from instructional demands, the practitioner increases the price of the break by requiring more behavior (and by extension, by increasing the delay to reinforcer receipt). If a reinforcer that maintains the most behavior amidst increasing prices is the best for producing behavior change, the first step will involve finding this reinforcer.

Applied behavior analysts typically attempt to identify this optimal "product" by conducting preference assessments. However, as demonstrated empirically by Roane, Lerman, and Vorndran (2001), a preference assessment may provide inadequate information about sensitivity to price increases. Under single-schedule conditions, Roane et al. progressively increased the price of stimuli that were equally and highly ranked based on the results of a paired-stimulus preference assessment (Fisher et al., 1992). Interestingly, all participants worked at least twice as hard to obtain one of the reinforcers despite their previous equal ranking. Because preference between two items that are virtually given away (in most preference assessments, the individual need only point to the item to receive it) tells us nothing about elasticity of demand, it is entirely possible that the highest price a client will pay for two different reinforcers would be the same, but one reinforcer might be ranked higher than the other on a preference assessment (the opposite of the Roane et al. finding). Results reported by Francisco, Borrero, and Sy (2008), while not definitive, are suggestive of this possibility (at the aggregate level). The results of the Roane et al. study suggest that determining how sensitive is an individual's demand for a reinforcer (i.e., quantifying elasticity of demand) may help the applied behavior analyst predict which consequence would produce greater response persistence in the applied setting in which the reinforcer will be used (Fisher & Mazur, 2001).

This position is elegantly outlined by the behavioral economists Hursh and Silberberg (2008). They assert that sensitivity to price changes (elasticity) is a better measure of reinforcer efficacy than more traditional measures, which can be affected by variables other than the reinforcer. For example, response rate can be affected both by the reinforcer and by the schedule of reinforcement that controls the delivery of that reinforcer. According to Hursh and Silberberg, the rate at which an individual responds to obtain an item is less important than how much he/she will continue to respond (and consume) in the face of continued price increases. Determining full demand curves is, no doubt, impractical in the context of a preference assessment (however, see Delmendo, Borrero, Beauchamp, & Francisco, in press, for one translational example). However, the preceding discussion suggests there may be merit in including elasticity probes within a preference assessment. This could be accomplished by identifying two or three standard tasks that could be completed with minimal instructions and would be rated by most as either easy, moderately effortful, or very effortful. By assessing demand for a reinforcer when it is delivered

contingent upon completing the individual tasks in turn, the researcher may better discriminate between reinforcers along the dimension of behavior-maintenance efficacy. As noted above, mapping these data on a demand curve can yield valuable information as to how we might expect behavioral output and reinforcer consumption to change at different prices. We feel that conceptualizing the response-reinforcer relationship as Hursh and Silberberg suggest – as an exchange between labor and goods available at varying prices – more closely simulates how reinforcers are earned in the natural environment. Therefore, if a reinforcer evaluation is conducted in a controlled setting (e.g., a room in which a preference assessment is normally conducted), one that is price based might allow for an intervention's smoother transition into the natural setting.

Substitutes and complements. Inelastic demand for a reinforcer in a controlled setting may not reflect sensitivity to price increases in a setting where other commodities are available. Consistent with Herrnstein's (1970) matching law, it is important for clinicians to not only understand how price changes affect demand for a reinforcer, but to also know how demand is influenced by the concurrent availability of other reinforcers. Behavioral economists have discussed a continuum of alternative reinforcers that can affect demand and consumer spending. At the ends of this continuum are perfect substitutes and complements.

A substitute reinforcer, as the name implies, is one that is functionally similar and is readily traded for the other reinforcer (Green & Freed, 1993). Common examples of substitutes include nickels and dimes, staples and paper clips, 7up® and Sprite®, and eyeglasses and contact lenses. When the price of one reinforcer increases, its consumption will decrease more if a substitute is available (Green & Rachlin, 1991). In other words, the availability of a substitute renders demand for a reinforcer more elastic than it would have been if the substitute were unavailable. This may translate to decreased treatment effectiveness. For example, if a therapist delivers sips of 7up® contingent on appropriate behavior, but the client receives free access to Sprite® throughout the day, demand for 7up® will be more elastic than it would be if response-contingent 7up® was the only carbonated lemon-lime beverage available. To obtain maximum amounts of appropriate behavior, the therapist will do well to either eliminate substitutes or find a new reinforcer that has no concurrently available substitute (this latter option may hold the most promise in applied contexts).

At the other end of the continuum of concurrently available reinforcers are complements. Unlike substitutes, two reinforcers that are classified as complements are those that are typically purchased and consumed together. For example, food and water, toothbrushes and toothpaste, and left and right shoes tend to be purchased and "consumed" together. Importantly, if consumption of one reinforcer (e.g., potato chips) declines when its price is increased, then consumption of a complement (e.g., chip dip) will also decline even though the latter has not been subject to a price increase. It may be important to know if a reinforcer to be used in a therapeutic setting is in a complementary relation with other reinforcers in the natural setting. For example, if a client is asked to choose between a ball and a book, the book may be preferred in the preference assessment context because reading is a solitary activity. However, the ball may be preferred and will maintain more behavior in the natural setting if complementary reinforcers are freely available (e.g., another person, a basketball hoop). Therefore, consideration of the context in which reinforcers will be consumed may impact their relative reinforcing efficacy (or value).

As argued above, assessing elasticity of demand for a reinforcer may help to identify more effective consequences to be used in applied settings. Important to note, however, is that behavior is rarely influenced by a single reinforcer in these settings. The presence of substitutes and complements may alter demand for the reinforcer from what was observed in a controlled setting. Thus, substitutes that compete with therapeutic reinforcers will diminish reinforcement-based intervention efficacy and they

should be minimized wherever possible. By contrast, concurrently available complementary reinforcers can increase consumption of the therapeutic reinforcer which, in turn, increases the frequency of the behavior targeted for improvement. Arranging complementary reinforcers may also render demand for the therapeutic reinforcer more inelastic. An understanding and consideration of the effects of price, substitutes, and complements may yield more effective and durable behavioral interventions.

Conclusions

The integration of traditional economic concepts within a science of behavior is still a relatively new venture but one that is gaining increased attention. Noneconomic behavioral approaches to therapeutic behavior change have proven their efficacy in affecting socially important behavior. We hope that this paper will promote the further incorporation of behavioral economic concepts into applied behavioral interventions. Stokes and Baer (1977) recognized the importance of maintaining behavioral improvements over time and generalizing these effects to different persons and settings. An understanding of how behavior is likely to change when the price of a reinforcer is increased and when substitutes and complements are concurrently available is a step in the direction of a systematic approach to generalization.

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